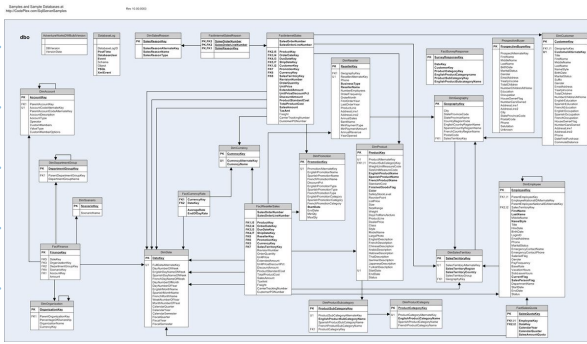


Join statements - 24/10/24

AdventureWorks 2008 Data Warehouse Schema



- we have multiple tables in datawarehouses
- this means our tables contain different information & they are related to one another
- Datawarehouse can also be called relational database & central database
- a place where we store data in tables
- a datawarehouse has multiple databases

DMS Data Management System (e.g. Databricks)

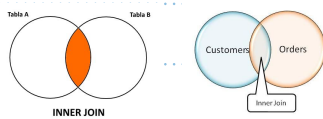
SQL Coding

We can join multiple tables - into one table

SQL Joins

- An SQL Join is meant to join multiple tables based on common columns.

INNER JOIN



- We join tables on columns but
- the join affects the output rows
- When we join tables, we can select
- any column between the tables we are joining

- Inner join returns rows where there is a match on both the tables

Syntax:

```
SELECT A.column 1,
       A.column 2,
       B.column 3,
       B.column 4
FROM table 1 AS A
INNER JOIN table 2 AS B
ON A.column - column = B.column - column
```

Remember the AS function is to rename

customers

cust_id	name
1	Rothi
2	Mbali

orders

order_id	cust_id	amount
101	1	500
102	4	20
103	5	10

We are going to join our table on the cust_id column

```
SELECT cust_id,
       name,
       amount,
FROM customers AS A
INNER JOIN orders AS B
ON A.cust_id = B.cust_id;
```



cust_id	name	amount
1	Rothi	500

Inner joins ensures that the outcome is what is appearing on both tables. It returns the rows where there is a match on the column we are joining on

• Primary key vs Foreign key

customers

the cust_id is different for each customer

primary key in the customers table

cust_id	name
1	Rothi
2	Mbali

orders

Primary key

foreign key in the orders table is not unique, it can be repeated

order_id	cust_id	amount
101	1	500
102	4	20
103	5	10
104	1	300

e.g.: you can have two or more separate orders but your cust_id remains the same

SELECT cust_id,
name,
amount,
FROM customers as A
INNER JOIN orders as B
ON A.cust_id = B.cust_id;

• A Primary Key is a unique identifier that identifies something uniquely (e.g. ID nr.)

• A Foreign Key is a Primary Key in one table and a Foreign Key in the other

cust_id	name	amount
1	Rothi	500
1	Rothi	300

• Dimension table vs Fact table

• Dimension table contains information that is descriptive (e.g. product names)

↳ Static table that contains info that is descriptive

• Fact table contains transactional info (e.g. revenue, amount) & it contains multiple keys

↳ Numerical table